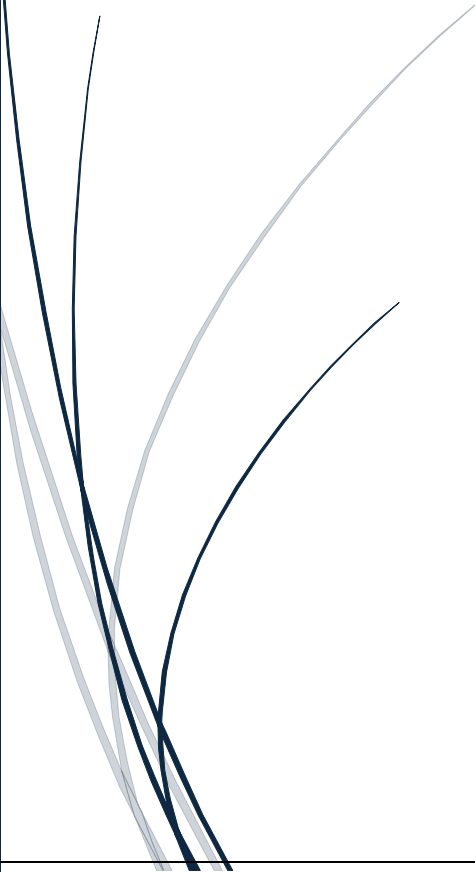




12/22/2025

E_Commerce_Store

Database project



Ahmed Hany Samir Ahmed ID:222400013
(SOLO PROJECT)

Project Title: E-Commerce Store Database System

Student Name: Ahmed Hany Samir Ahmed

Student ID: 222400013

Submission Date: 12/22/2025

My Contributions to the Project:

As the project's sole developer, I conceptualized, designed, and implemented the complete database system. My comprehensive involvement spanned all project phases:

1. *Project Requirements Analysis & Planning*
 - *Analyzed business needs for an e-commerce store*
 - *Identified core entities and relationships*
 - *Defined project scope and objectives*
2. *Database Conceptual Design*
 - *Created Entity-Relationship (ER) diagram with proper cardinality*
 - *Documented all entities, attributes, and relationships*
 - *Defined business rules and constraints*
3. *Logical & Physical Database Design*
 - *Designed relational schema with normalized tables*
 - *Selected appropriate data types and lengths for all attributes*
 - *Defined primary keys, foreign keys, and indexes*
4. *Database Implementation*
 - *Wrote all SQL CREATE TABLE statements*
 - *Implemented all foreign key constraints using ALTER TABLE*
 - *Added data integrity constraints (CHECK constraints)*
 - *Created and populated the database with comprehensive sample data*
5. *Application Design & Documentation*
 - *Designed the PHP web application structure*
 - *Documented file organization and functionality*
 - *Planned user interfaces for both customer and admin roles*
6. *Testing & Validation*
 - *Tested all SQL statements for syntax and logic*
 - *Validated foreign key relationships and constraints*
 - *Ensured data consistency across all tables*
7. *Report Writing & Documentation*
 - *Wrote the complete project report including all required sections*
 - *Created professional documentation with proper formatting*
 - *Prepared additional data files for extended testing scenarios*

1. The Main Purpose of the Project

Project Goals and Business Description

The main goal of this project is to analyze, design, and implement a robust, scalable, and secure **Database Management System (DBMS)** to support a full-featured online retail operation (E-commerce Store)⁷. The application will be a web-based storefront and administrative backend, facilitating the sale of various products across multiple categories.

Problem/Opportunity Being Addressed

The project addresses the need for a **single, reliable source of truth** for all core business data. Without a structured database, the business would suffer from inaccurate inventory, manual and error-prone order processing, and inability to track customer behavior or manage promotions effectively. The opportunity is to leverage a relational database to:

Maintain accurate Inventory: Track stock levels for products and their specific **variants** (e.g., size and color)⁸.

Streamline Order Processing: Record orders, payment details, shipping addresses, and apply promotions.

Support Marketing: Manage coupons and promotions to drive sales.

Enhance User Experience: Provide customers with secure accounts, order history, and saved addresses.

Identification of Information Needs

The information needed to solve these problems and manage the business falls into four primary domains:

1. **User Management:** Full name, unique email, phone number, secure password, user role (customer/admin), and multiple saved shipping addresses.
2. **Product Management:** Product name, detailed description, base price, category association, specific product variants (size, color, etc.), stock quantity, and product availability status.
3. **Transaction Processing:** Order date, current order status (pending, shipped, etc.), total amount, payment method, and the specific items (with quantity and price) included in each order.
4. **Marketing & Promotions:** Coupon codes, discount value, validity dates, and minimum order requirements.

Entity and Attribute Details

This table lists every attribute for each entity (table) in the schema, including technical constraints (data type, length)

Table	Attribute	Data Type	Length/Precision	Nullable	Unique	Default	Description	Business Constraints
Users	user_id	INT	10	NO	Auto	Auto	Unique user identifier	Primary Key, Auto-increment
	email	VARCHAR	255	NO	YES	-	User email address	Unique
	password	VARCHAR	255	NO	NO	-	Plain text password	-
	full_name	VARCHAR	255	NO	NO	-	User's full name	Non-empty string
	phone_number	VARCHAR	255	NO	YES	-	Contact number	Unique
	created_at	TIMESTAMP	-	NO	NO	Current	Account creation	Auto-set on creation
	role	VARCHAR	255	YES	NO	-	User role	customer', 'admin', 'moderator'
User_Addresses	address_id	INT	10	NO	Auto	Auto	Unique address ID	Primary Key, Auto-increment
	user_id	INT	10	NO	NO	-	Reference to user	FK to Users
	label	VARCHAR	255	NO	NO	-	Label (Home/Work)	Descriptive label
	street_address	VARCHAR	255	NO	NO	-	Street address	-
	city	VARCHAR	255	NO	NO	-	City name	-
	zip_code	INT	10	YES	NO	-	Postal code	-
Products	product_id	INT	10	NO	Auto	Auto	Unique product ID	Primary Key, Auto-increment
	name	VARCHAR	255	NO	NO	-	Product name	Descriptive, non-empty
	description	VARCHAR	255	YES	NO	-	Product description	Optional details
	price	DECIMAL	10,2	NO	NO	-	Product price	≥ 0
	category_id	INT	10	NO	NO	-	Product category	FK to Categories
	status	VARCHAR	255	YES	NO	-	Availability	active', 'inactive', 'out_of_stock'
	image_url	VARCHAR	255	YES	NO	-	Product image URL	Valid URL format
Categories	category_id	INT	10	NO	Auto	Auto	Unique identifier	Primary Key, Auto-increment
	name	VARCHAR	255	NO	YES	-	Category name	Must be unique
Product_Variants	variant_id	INT	10	NO	Auto	Auto	Unique variant ID	Primary Key, Auto-increment
	product_id	INT	10	NO	NO	-	Parent product	FK to Products
	stock_quantity	INT	10	YES	NO	0	Available stock	≥ 0
	is_available	BIT	1	YES	NO	1	Availability flag	1=Available, 0=Unavailable
Variant_Attributes	variant_attribute_id	INT	10	NO	Auto	Auto	Unique attribute ID	Primary Key, Auto-increment
	variant_id	INT	10	NO	NO	-	Associated variant	FK to Product_Variants
	attribute_name	VARCHAR	50	NO	NO	-	Attribute type	e.g., 'Color', 'Size'

	attribute_value	VARCHAR	50	YES	NO	-	Attribute value	e.g., 'Red', 'XL'
Orders	order_id	INT	10	NO	Auto	Auto	Unique order ID	Primary Key, Auto-increment
	user_id	INT	10	NO	NO	-	Customer	FK to Users
	order_date	TIMESTAMP	-	NO	NO	Current	Order timestamp	Auto-set on creation
	status	VARCHAR	255	YES	NO	-	Order status	pending', 'processing', 'shipped'
	coupon_id	INT	10	YES	NO	-	Applied coupon	FK to Promotions_and_Coupons
	discount_applied	DECIMAL	10,2	YES	NO	0	Discount amount	0 ≤ value ≤ subtotal
	address_id	INT	10	NO	NO	-	Shipping address	FK to User_Addresses
	payment_method	VARCHAR	50	YES	NO	-	Payment type	e.g., 'Credit Card', 'PayPal'
	final_total	DECIMAL	10,2	NO	NO	-	Final amount paid	≥ 0
	shipping_cost	DECIMAL	10,2	NO	NO	-	Shipping charges	≥ 0
Order_Items	order_item_id	INT	10	NO	Auto	Auto	Unique item ID	Primary Key, Auto-increment
	order_id	INT	10	NO	NO	-	Parent order	FK to Orders
	quantity	INT	10	NO	NO	1	Item quantity	> 0
	unit_price	DECIMAL	10,2	NO	NO	-	Price at purchase	≥ 0
	variant_id	INT	10	NO	NO	-	Product variant	FK to Product_Variants
Shopping_Cart	cart_item_id	INT	10	NO	Auto	Auto	Unique cart item ID	Primary Key, Auto-increment
	quantity	INT	10	NO	NO	1	Item quantity	> 0, ≤ available stock
	user_id	INT	10	NO	NO	-	Cart owner	FK to Users
	variant_id	INT	10	NO	NO	-	Selected variant	FK to Product_Variants
Payments	payment_id	INT	10	NO	Auto	Auto	Unique payment ID	Primary Key, Auto-increment
	order_id	INT	10	NO	YES	-	Associated order	FK to Orders, Unique
	payment_date	TIMESTAMP	-	NO	NO	Current	Payment timestamp	Auto-set on payment
	amount	DECIMAL	10,2	NO	NO	-	Payment amount	> 0, = order.final_total
Promotions_and_Coupons	coupon_id	INT	10	NO	Auto	Auto	Unique coupon identifier	Primary Key, Auto-increment
	code	VARCHAR	255	YES	YES	-	Coupon code	Unique, can be null
	discount_value	DECIMAL	10,2	YES	NO	-	Discount amount	Positive value expected
	valid_from	DATE	-	YES	NO	-	Activation date	Must be ≤ expires_at
	expires_at	DATE	-	YES	NO	-	Expiration date	Must be ≥ valid_from
	min_order_total	DECIMAL	10,2	YES	NO	-	Minimum order total	Positive value if provided

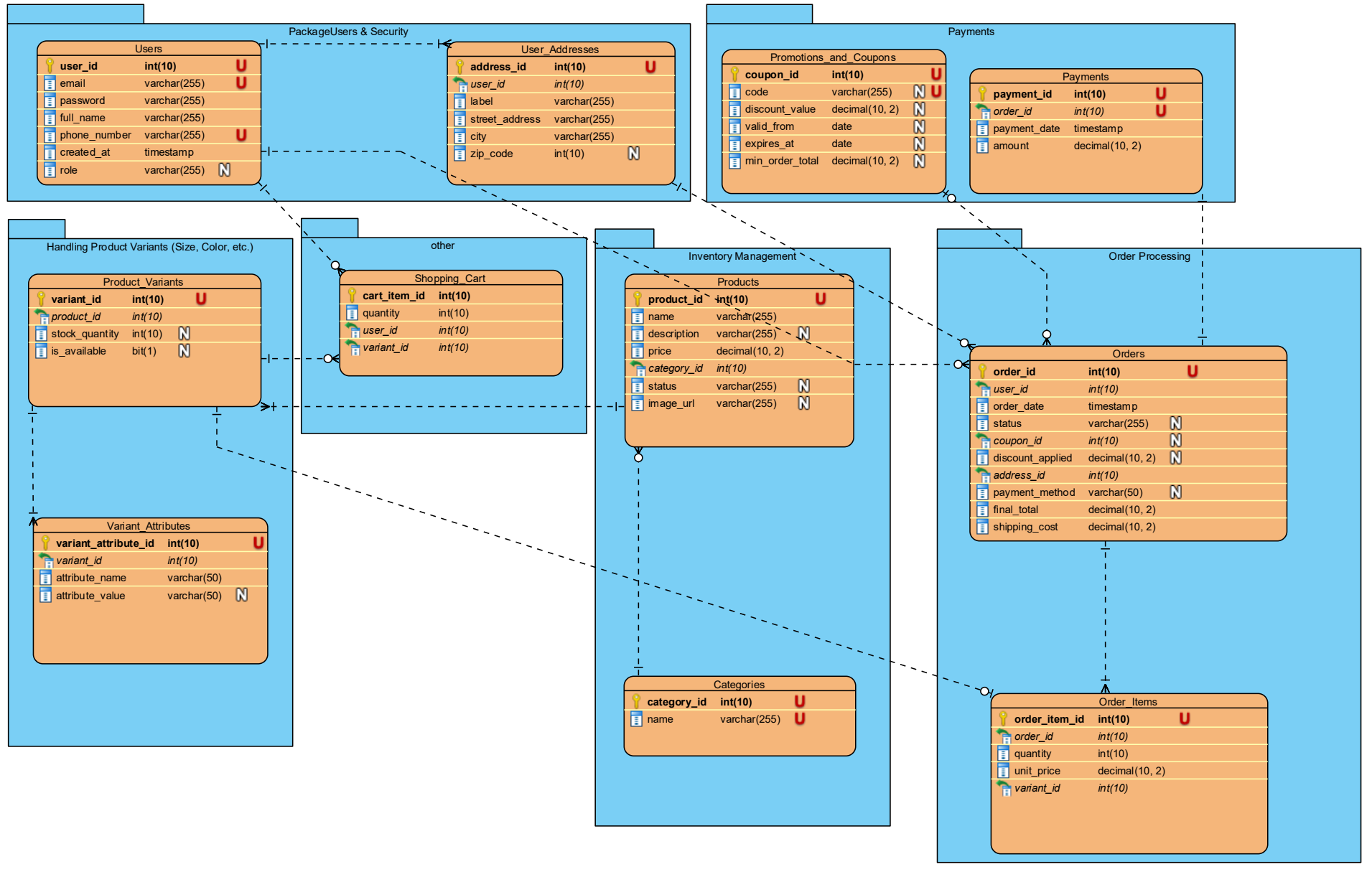
Relationships Between Entities

This table lists every attribute for each entity (table) in the schema, including technical constraints (data type, length) and business constraints (rules/checks).

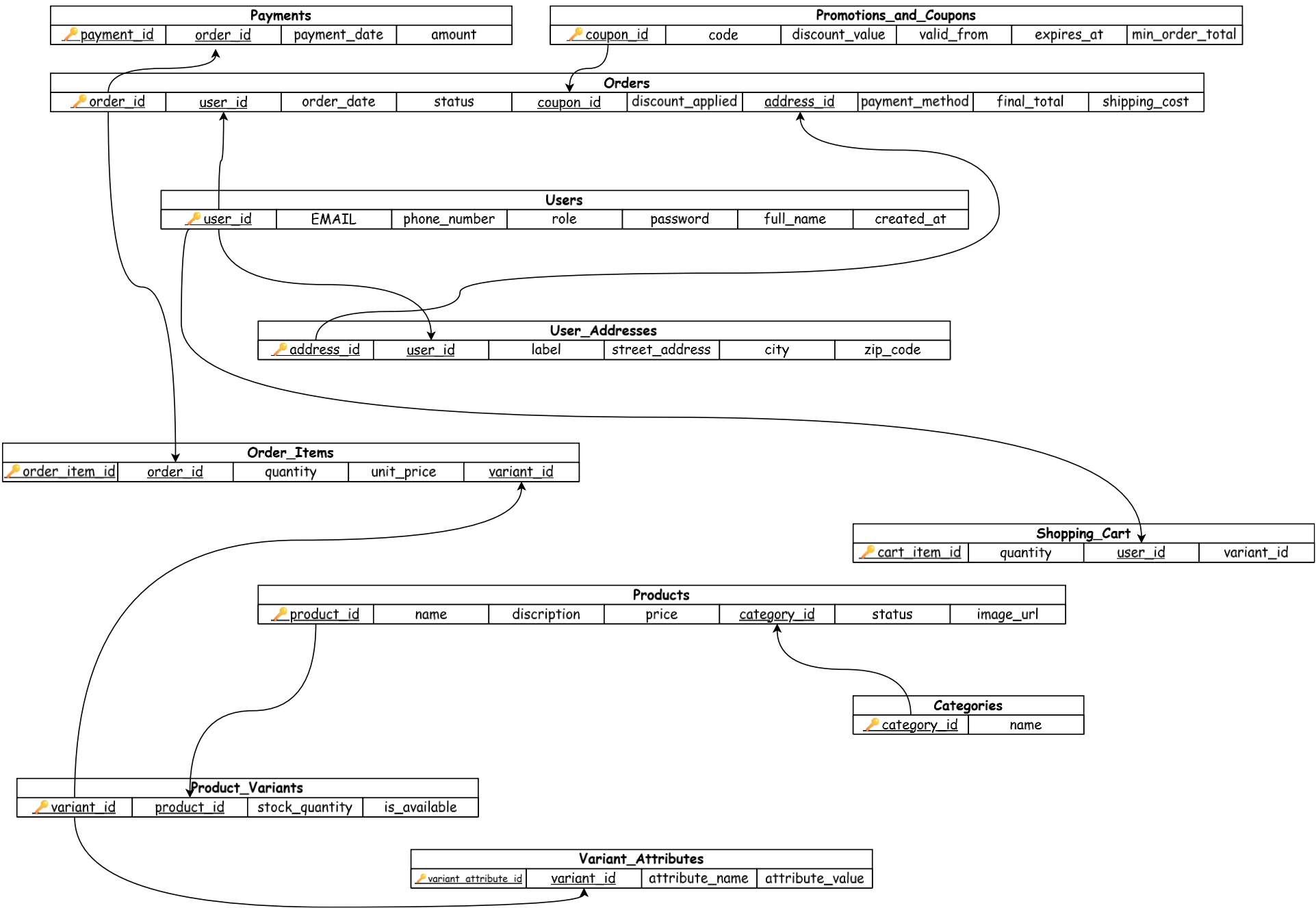
RELATIONSHIP NAME	RELATIONSHIP DESCRIPTION (PARTICIPATION : CARDINALITY)	FK COLUMN	LOGIC
USER ORDERS	User (0:N) has Order (1:1)	Orders.user_id	A User can have zero or many orders. Each Order must belong to exactly one User.
USER ADDRESSES	User (0:N) has Address (1:1)	User_Addresses.user_id	A user can have zero or many addresses. Each address must belong to one user.
USER CART	User (0:N) has Cart_Item (1:1)	Shopping_Cart.user_id	A user can have zero or many items in cart. Each item must have an owner.
ORDER SHIPPING	Address (0:N) has Order (1:1)	Orders.address_id	An address can be used for many orders. Each Order must have one address.
ORDER ITEMS	Order (1:N) has Order_Item (1:1)	Order_Items.order_id	An Order must have at least one item. Each item must belong to one order.
ORDER PAYMENT	Order (1:1) has Payment (1:1)	Payments.order_id	An Order must have one payment. Each payment must link to one order.
ORDER COUPON	Coupon (0:N) has Order (0:1)	Orders.coupon_id	A coupon can be on many orders. An order can have zero or one coupon.
PRODUCT CATEGORY	Category (0:N) has Product (1:1)	Products.category_id	A category can have many products. Each Product must have a category.
PRODUCT VARIANTS	Product (1:N) has Variant (1:1)	Product_Variants.product_id	A product must have variants. Each variant must link to one product.
VARIANT ATTRIBUTES	Variant (1:N) has Attribute (1:1)	Variant_Attributes.variant_id	A variant must have attributes. Each attribute must link to one variant.
CART VARIANT	Variant (0:N) has Cart_Item (1:1)	Shopping_Cart.variant_id	A variant can be in many carts. Each cart item must link to one variant.
ITEM VARIANT	Variant (0:N) has Order_Item (1:1)	Order_Items.variant_id	A variant can be in many orders. Each order item must link to one variant.

ER diagram

E-Commerce Store



Relational Schema Mapping





Database and Schema Setup

--Create the Database

```
CREATE DATABASE E_Commerce_Store;
```

```
USE E_Commerce_Store;
```



Table Creation

-- Category Table

```
CREATE TABLE Categories (  
  category_id int(10) NOT NULL AUTO_INCREMENT,  
  name        varchar(255) NOT NULL UNIQUE,  
  PRIMARY KEY (category_id));
```

-- Promotions and Coupons Table

```
CREATE TABLE Promotions_and_Coupons (  
  coupon_id      int(10) NOT NULL AUTO_INCREMENT,  
  code           varchar(255) UNIQUE,  
  discount_value decimal(10, 2),  
  valid_from     date,  
  expires_at     date,  
  min_order_total decimal(10, 2),  
  PRIMARY KEY (coupon_id));
```

-- Users Table

```
CREATE TABLE Users (  
  user_id      int(10) NOT NULL AUTO_INCREMENT,  
  email        varchar(255) NOT NULL UNIQUE,  
  password     varchar(255) NOT NULL,  
  full_name    varchar(255) NOT NULL,  
  phone_number varchar(255) NOT NULL UNIQUE,  
  created_at   timestamp NOT NULL,  
  role         varchar(255),  
  PRIMARY KEY (user_id));
```

-- User Addresses

```
CREATE TABLE User_Addresses (  
  address_id    int(10) NOT NULL AUTO_INCREMENT,  
  user_id       int(10) NOT NULL,  
  label         varchar(255) NOT NULL,  
  street_address varchar(255) NOT NULL,  
  city          varchar(255) NOT NULL,  
  zip_code      int(10),  
  PRIMARY KEY (address_id));
```

-- Product Definition

```
CREATE TABLE Products (  
    product_id int(10) NOT NULL AUTO_INCREMENT,  
    name        varchar(255) NOT NULL,  
    description varchar(255),  
    price       decimal(10, 2) NOT NULL,  
    category_id int(10) NOT NULL,  
    status       varchar(255),  
    image_url    varchar(255),  
    PRIMARY KEY (product_id));
```

-- Product Variants

```
CREATE TABLE Product_Variants (  
    variant_id int(10) NOT NULL AUTO_INCREMENT,  
    product_id int(10) NOT NULL,  
    stock_quantity int(10),  
    is_available bit(1),  
    PRIMARY KEY (variant_id));
```

-- Variant Attributes (Color, Size, etc.)

```
CREATE TABLE Variant_Attributes (  
    variant_attribute_id int(10) NOT NULL AUTO_INCREMENT,  
    variant_id           int(10) NOT NULL,  
    attribute_name        varchar(50) NOT NULL,  
    attribute_value        varchar(50),  
    PRIMARY KEY (variant_attribute_id));
```

-- Shopping Cart Items

```
CREATE TABLE Shopping_Cart (  
    cart_item_id int(10) NOT NULL AUTO_INCREMENT,  
    quantity      int(10) NOT NULL,  
    user_id       int(10) NOT NULL,  
    variant_id    int(10) NOT NULL,  
    PRIMARY KEY (cart_item_id));
```

-- Customer Orders

```
CREATE TABLE Orders (  
    order_id int(10) NOT NULL AUTO_INCREMENT,  
    user_id  int(10) NOT NULL,  
    order_date timestamp NOT NULL,  
    status    varchar(255),  
    coupon_id int(10),  
    discount_applied decimal(10, 2),  
    address_id int(10) NOT NULL,  
    payment_method varchar(50),  
    final_total decimal(10, 2) NOT NULL,  
    shipping_cost decimal(10, 2) NOT NULL,
```

```

    PRIMARY KEY (order_id));
-- Order Item Details
CREATE TABLE Order_Items (
    order_item_id int(10) NOT NULL AUTO_INCREMENT,
    order_id      int(10) NOT NULL,
    quantity      int(10) NOT NULL,
    unit_price     decimal(10, 2) NOT NULL,
    variant_id     int(10) NOT NULL,
    PRIMARY KEY (order_item_id));
-- Payment Records
CREATE TABLE Payments (
    payment_id    int(10) NOT NULL AUTO_INCREMENT,
    order_id      int(10) NOT NULL UNIQUE,
    payment_date  timestamp NOT NULL,
    amount        decimal(10, 2) NOT NULL,
    PRIMARY KEY (payment_id));
-- Shopping Cart Items
CREATE TABLE Shopping_Cart (
    cart_item_id int(10) NOT NULL AUTO_INCREMENT,
    quantity      int(10) NOT NULL,
    user_id       int(10) NOT NULL,
    variant_id     int(10) NOT NULL,
    PRIMARY KEY (cart_item_id));
-- Customer Orders
CREATE TABLE Orders (
    order_id      int(10) NOT NULL AUTO_INCREMENT,
    user_id       int(10) NOT NULL,
    order_date     timestamp NOT NULL,
    status        varchar(255),
    coupon_id      int(10),
    discount_applied decimal(10, 2),
    address_id     int(10) NOT NULL,
    payment_method varchar(50),
    final_total    decimal(10, 2) NOT NULL,
    shipping_cost  decimal(10, 2) NOT NULL,
    PRIMARY KEY (order_id));
-- Order Item Details
CREATE TABLE Order_Items (
    order_item_id int(10) NOT NULL AUTO_INCREMENT,
    order_id      int(10) NOT NULL,
    quantity      int(10) NOT NULL,
    unit_price     decimal(10, 2) NOT NULL,
    variant_id     int(10) NOT NULL,

```

```

PRIMARY KEY (order_item_id));
-- Payment Records
CREATE TABLE Payments (
  payment_id  int(10) NOT NULL AUTO_INCREMENT,
  order_id    int(10) NOT NULL UNIQUE,
  payment_date timestamp NOT NULL,
  amount      decimal(10, 2) NOT NULL,
  PRIMARY KEY (payment_id));

```

Foreign Key Constraints

```

-- Relationships with Users
ALTER TABLE Orders
ADD CONSTRAINT FKOrders145166
FOREIGN KEY (user_id) REFERENCES Users (user_id);
ALTER TABLE User_Addresses
ADD CONSTRAINT FKUser_Addre599603
FOREIGN KEY (user_id) REFERENCES Users (user_id);
ALTER TABLE Shopping_Cart
ADD CONSTRAINT FKShopping_C125642
FOREIGN KEY (user_id) REFERENCES Users (user_id);
-- Relationships for Products
ALTER TABLE Products
ADD CONSTRAINT FKProducts459619
FOREIGN KEY (category_id) REFERENCES Categories (category_id);
ALTER TABLE Product_Variants
ADD CONSTRAINT FKProduct_Va769126
FOREIGN KEY (product_id) REFERENCES Products (product_id);
ALTER TABLE Variant_Attributes
ADD CONSTRAINT FKVariant_At263396
FOREIGN KEY (variant_id) REFERENCES Product_Variants (variant_id);
ALTER TABLE Shopping_Cart
ADD CONSTRAINT FKShopping_C607982
FOREIGN KEY (variant_id) REFERENCES Product_Variants (variant_id);
-- Relationships for Orders and Payments
ALTER TABLE Orders
ADD CONSTRAINT FKOrders151327
FOREIGN KEY (address_id) REFERENCES User_Addresses (address_id);
ALTER TABLE Orders
ADD CONSTRAINT FKOrders643118
FOREIGN KEY (coupon_id) REFERENCES Promotions_and_Coupons (coupon_id);
ALTER TABLE Order_Items
ADD CONSTRAINT FKOrder_Item921064
FOREIGN KEY (order_id) REFERENCES Orders (order_id);

```

```

ALTER TABLE Order_Items
ADD CONSTRAINT FKOrder_Item350744
FOREIGN KEY (variant_id) REFERENCES Product_Variants (variant_id);

ALTER TABLE Payments
ADD CONSTRAINT FKPayments35249
FOREIGN KEY (order_id) REFERENCES Orders (order_id);

```

Data Integrity Constraints

```

-- User Constraints
ALTER TABLE Users ADD CONSTRAINT chk_user_role
CHECK (role IN ('customer', 'admin', 'moderator'));

-- Product Constraints
ALTER TABLE Products ADD CONSTRAINT chk_product_status
CHECK (status IN ('active', 'inactive', 'out_of_stock'));
ALTER TABLE Products ADD CONSTRAINT chk_price_positive
CHECK (price >= 0);

-- Inventory Constraints
ALTER TABLE Product_Variants ADD CONSTRAINT chk_stock_non_negative
CHECK (stock_quantity >= 0);

-- Order and Transaction Constraints
ALTER TABLE Orders ADD CONSTRAINT chk_order_status
CHECK (status IN ('pending', 'processing', 'shipped', 'delivered', 'cancelled'));
ALTER TABLE Order_Items ADD CONSTRAINT chk_quantity_positive
CHECK (quantity > 0);

```

Random data added to database

```

-- 1. BASIC USERS (5 users)
INSERT INTO Users (user_id, email, password, full_name, phone_number, created_at, role) VALUES
(1, 'admin@store.com', 'admin123', 'Admin User', '1110000000', NOW(), 'admin'),
(2, 'customer@test.com', 'customer123', 'Test Customer', '2221111111', NOW(), 'customer'),
(3, 'user1@test.com', 'password123', 'User One', '3332222222', NOW(), 'customer'),
(4, 'user2@test.com', 'password123', 'User Two', '4443333333', NOW(), 'customer'),
(5, 'user3@test.com', 'password123', 'User Three', '5554444444', NOW(), 'customer');

-- 2. SIMPLE CATEGORIES (3 categories)
INSERT INTO Categories (category_id, name) VALUES
(1, 'Electronics'),
(2, 'Clothing'),
(3, 'Books');

-- 3. BASIC PRODUCTS (5 products)
INSERT INTO Products (product_id, name, description, price, category_id, status, image_url) VALUES

```

```

(1, 'Laptop', 'High-performance laptop', 1000.00, 1, 'active', 'url/products/laptop.jpg'),
(2, 'T-Shirt', 'Cotton t-shirt', 25.00, 2, 'active', 'url/products/tshirt.jpg'),
(3, 'Headphones', 'Wireless headphones', 100.00, 1, 'active', 'url/products/headphones.jpg'),
(4, 'Programming Book', 'Learn to code', 50.00, 3, 'active', 'url/products/book.jpg'),
(5, 'Jeans', 'Denim jeans', 60.00, 2, 'active', 'url/products/jeans.jpg');

-- 4. PRODUCT VARIANTS (1-2 variants per product)
INSERT INTO Product_Variants (variant_id, product_id, stock_quantity, is_available) VALUES
(1, 1, 10, 1), -- Laptop variant
(2, 2, 20, 1), (3, 2, 15, 1), -- T-Shirt variants
(4, 3, 30, 1), -- Headphones
(5, 4, 25, 1), -- Book
(6, 5, 10, 1), (7, 5, 5, 1); -- Jeans variants

-- 5. VARIANT ATTRIBUTES (simple attributes)
INSERT INTO Variant_Attributes (variant_id, attribute_name, attribute_value) VALUES
(1, 'RAM', '16GB'),
(2, 'Size', 'M'), (3, 'Size', 'L'),
(4, 'Color', 'Black'),
(5, 'Format', 'Paperback'),
(6, 'Waist', '32'), (7, 'Waist', '34');

-- 6. USER ADDRESSES (2 addresses)
INSERT INTO User_Addresses (address_id, user_id, label, street_address, city, zip_code) VALUES
(1, 2, 'Home', '123 Main St', 'Test City', 12345),
(2, 3, 'Office', '456 Office Ave', 'Work City', 67890);

-- 7. COUPONS (3 coupons)
INSERT INTO Promotions_and_Coupons (coupon_id, code, discount_value, valid_from, expires_at, min_order_total) VALUES
(1, 'WELCOME10', 10.00, '2025-01-01', '2025-12-31', 50.00),
(2, 'SAVE5', 5.00, '2025-01-01', '2025-12-31', 25.00),
(3, 'FREESHIP', 0.00, '2025-01-01', '2026-01-01', 30.00);

-- 8. ORDERS (3 sample orders)
INSERT INTO Orders (order_id, user_id, order_date, status, coupon_id, discount_applied, address_id, payment_method, final_total, shipping_cost) VALUES
(1, 2, DATE_SUB(NOW(), INTERVAL 7 DAY), 'delivered', 1, 10.00, 1, 'Credit Card', 990.00, 5.00),
(2, 3, DATE_SUB(NOW(), INTERVAL 3 DAY), 'processing', NULL, 0.00, 2, 'PayPal', 150.00, 5.00),
(3, 2, NOW(), 'pending', 2, 5.00, 1, 'Credit Card', 80.00, 5.00);

-- 9. ORDER ITEMS (few items)
INSERT INTO Order_Items (order_item_id, order_id, quantity, unit_price, variant_id) VALUES
(1, 1, 1, 1000.00, 1), -- Laptop
(2, 2, 2, 25.00, 2), -- 2 T-Shirts
(3, 2, 1, 100.00, 4), -- Headphones
(4, 3, 1, 50.00, 5); -- Book

```

```
-- 10. SHOPPING CART ITEMS (for active users)
INSERT INTO Shopping_Cart (cart_item_id, quantity, user_id, variant_id) VALUES
(1, 1, 2, 6), -- Jeans in cart
(2, 2, 3, 3), -- 2 T-Shirts in cart
(3, 1, 4, 4); -- Headphones in cart

-- 11. PAYMENTS (simple payments)
INSERT INTO Payments (payment_id, order_id, payment_date, amount) VALUES
(1, 1, DATE_SUB(NOW(), INTERVAL 7 DAY), 990.00),
(2, 2, DATE_SUB(NOW(), INTERVAL 3 DAY), 150.00);

-- Re-enable foreign key checks
SET FOREIGN_KEY_CHECKS = 1;
```


5. {BONUS} Application Implementation

The application utilizes PHP for the front-end forms and management menus, and the database schema to generate necessary reports and complex queries.

```
ecommerce_project/
├── admin/ (12 files)
│   ├── categories.php      # Manage product categories (add/edit/delete)
│   ├── coupons.php         # Create/edit/delete discount coupons
│   ├── dashboard.php       # Admin homepage with stats and navigation
│   ├── manage_payments.php  # View payment reports and revenue analytics
│   ├── manage_user_addresses.php # Manage user shipping addresses
│   ├── manage_users.php     # CRUD operations for user accounts
│   ├── order_details.php    # Detailed view of a specific order
│   ├── order_edit_status.php # Update order status (pending/processing/shipped/delivered/cancelled)
│   ├── orders.php          # View all customer orders
│   ├── products.php        # Manage product listings (add/edit/delete)
│   ├── variant_attributes.php # Manage variant-specific attributes (color, size, etc.)
│   └── variants.php        # Manage product variants (stock, pricing, availability)
├── assets/ (2 files)
│   ├── css/
│   │   └── styles.css      # Main CSS stylesheet for entire site
│   └── images/
│       └── placeholder.png # Default placeholder image for products
├── auth/ (3 files)
│   ├── login.php          # User login with role-based redirect (admin/customer)
│   ├── logout.php         # Session termination and logout
│   └── register.php       # New user registration
├── cart/ (4 files)
│   ├── add_to_cart.php     # Add product variants to shopping cart
│   ├── cart.php           # View and manage cart items
│   ├── remove_cart.php    # Remove items from cart
│   └── update_cart.php     # Update item quantities in cart
├── config/ (1 file)
│   └── db.php             # Database connection configuration and setup
├── customer/ (8 files)
│   ├── checkout.php       # Checkout process with address selection AND coupon application
│   ├── manage_addresses.php # User's shipping address management (CRUD)
│   ├── order_details.php  # Customer's order details view
│   ├── order_success.php  # Order confirmation/success page
│   ├── orders.php         # Customer's order history with status AND cancellation ability
│   ├── payment.php        # Payment processing with coupon application AND simulated gateway
│   ├── products.php       # Product catalog with filtering and search
│   └── profile.php        # User profile management (update details, password)
├── includes/ (3 files)
│   ├── footer.php         # Site footer with scripts and copyright
│   ├── functions.php      # Utility functions (flash messages, validation, helpers) -ALSO contains set_flash_message(), get_flash_message(), display_flash_message()
│   └── header.php         # Site header with navigation and session handling
├── url/ (115 image files total)
│   └── products/          # Product images directory
│       ├── (40 main product images)
│       ├── (70 additional product images)
│       └── (5 testing images)
├── index.php              # Homepage with featured products and categories
└── product.php            # Individual product detail page with variants
```